Post-Disaster Damage Assessment and Rescue Guidance

CSE499B – Section 13, Group 5, Update 4

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Weekly Update:

Our system reads before-and-after satellite images of a disaster, marks the damage building by building, counts it, and writes a rescue report that cites official guidelines, offline, on hardware small enough for the field. This week's main result is the report-writing system's proof run: with the internet switched off, our own fine-tuned vision-language model (VLM, a Qwen2.5-VL-7B) wrote all six reports, and nothing left the machine. We also report the first version of our website and a second, larger test set. Segmentation scores are on our 645 validation image pairs, or, where marked clean, on the 605 left after removing 40 whose before-image also appears in training.

Results Summary

Goal	Target	Best result so far	Status
1. Damage map	Mean IoU above 0.52 and destroyed-building IoU above 0.30, on two training seeds	Find-then-grade, clean pairs: destroyed-building IoU 0.316, 0.305 and 0.298 with three graders (mean 0.307); mean IoU 0.547	Met on the mean of three graders, narrowly
2. Counting	Above the base VLM's 34.2% on the benchmark's counting questions	32.96% by a fixed query over a table of the buildings our U-Net found (32.47% without duplicated images)	1.7 points short
3. Offline report	The report-writing agents on our own models, with damage map, counts and rescue heatmap	A full run on our 7B with the internet switched off: six reports, no connection outside the machine	Offline proof done; report quality next
4. Field model	A small segmenter and a distilled 3B VLM	U-Net in 8-bit integers: 33 MB, 1.43× faster on a CPU, destroyed IoU down 0.0003; 1,146 training answers for the 3B	Segmenter done; 3B training next

IoU (intersection over union): the overlap between the pixels predicted as a label and the pixels truly of it, from 0 to 1; mean IoU averages it over background, intact, damaged and destroyed.

Ridita Afrin Riya – Report Agents on Our Model, Offline

1. **Agents or a fixed flow:** three more practice runs of DisasTeller's four report agents on our 7B, each failing differently: an agent repeated the same tool call until one message passed the model's 16,384-token limit; agents wrote one sentence about the report instead of the report; agents repeated a call until their step limit, which saved the call itself as the report. By a rule she wrote before the third run, she switched to a fixed flow: code calls the photo tool, the map tool and 18 guideline searches in a fixed order, and the 7B writes each report section by section (28 sections).
2. **The proof run, internet off:** it passed all five rules fixed before any run: the run finished; six reports of over 50 words (192 to 787 written by the model); no connection outside the machine (four, all to the model server on the same machine); no Gemini library loaded; every model call answered. Its reports match the internet-on run word for word (6 minutes on two T4 GPUs). It does not show that the reports are good: the map reading says there are no dangerous zones, which three reports repeat, one report invents a hotline number, and only one cites guideline sections. These go into the comparison with the reports Gemini wrote in July.

Abdullah Al Noman – Showcase Website

1. **A first version of the website:** seven pages built with React for Vercel (home, data, method, results, gallery, demo, about). The demo page sends a before-and-after pair to a local server and labels its output as a sample until one is connected, and every page carries the decision-support warning. Its results pages are being checked against our results files.

Abrar Mohammed Tanzim Alam – A Second Test Set

1. **Test pairs no model has seen:** our 645 validation pairs were also used to choose between models. The benchmark's own test split has 2,067 optical before-and-after pairs, with one damage mask per class. We merged them by the rule our training used into one map per pair (background, intact, damaged, destroyed), checked every mask against its image, and left out 60 pairs: 2 with a missing file, 1 failing the mask check, and 57 whose image also appears in our training data. That leaves 2,007 clean test pairs for checking our models on unseen images.

Next Week

- **Damage map:** compare R11 and find-then-grade on the new test pairs, and choose the one that feeds the report.
- **Field model:** train the 3B on the 7B's answers.
- **Offline report:** compare the six offline reports with Gemini's, then fix the map reading and the citations, as a new run.
- **Website:** correct its results pages against our results files.